

Advanced (Habanero)

Q1. Solve for x : $2^x = 8$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Solve for y : $3^y = 27$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q3. Solve for z : $4^z = 16$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q4. Solve for x : $x^2 = 25$

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = pm5$

Q5. Solve for y : $y^3 = 64$

- (A) $y = 2$
- (B) $y = 3$
- (C) $y = 4$
- (D) $y = 5$
- (E) $y = 6$

Q6. Solve for z : $z^4 = 81$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = pm3$

Q7. Solve for x : $2^x = 32$

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

Q8. Solve for y : $5^y = 125$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Open Ended Questions (FRQ)

Q9. Solve for x : $x^2 - 4x - 8 = 0$

Q10. Solve for y : $y^3 - 2y^2 - 15y = 0$

Chef's Tips

- Tip 1: Make sure to check for extraneous solutions
- Tip 2: Use properties of exponents to simplify equations
- Tip 3: Graphing calculators are not allowed